Akshay Ballal

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ABOUT ME

Versatile AI Engineer with a background in developing deep learning models and using AI in products. I have authored two open source libraries, out of which one has over 200,000 downloads and over 500 stars on GitHub. I am pursuing an MSc in Artificial Intelligence and Engineering Systems at Eindhoven University of Technology. I have extensive expertise in machine learning, software development, and engineering.

EDUCATION AND TRAINING

04/09/2023 – CURRENT Eindhoven, Netherlands
MSC. ARTIFICIAL INTELLIGENCE AND ENGINEERING SYSTEMS Eindhoven University of Technology

Website https://www.tue.nl/en/

03/08/2013 – 03/06/2017 Ranchi, India
BACHELOR OF ENGINEERING (MECHANICAL) Birla Institute of Technology, Mesra

Website www.bitmesra.ac.in

WORK EXPERIENCE

ASML – VELDHOVEN, NETHERLANDS

MACHINE LEARNING INTERN – 04/11/2024 – CURRENT

- **Developed denoising diffusion models** for unsupervised detection of process drift in the lithography pipeline, enabling early-stage anomaly detection without labeled data.
- **Engineered a novel variant of the Variational Autoencoder (VAE)** to extract meaningful Key Performance Indicators (KPIs) from high-dimensional optical metrology data, significantly reducing dependence on costly Scanning Electron Microscope (SEM) measurements.
- **Integrated uncertainty quantification techniques** to generate confidence intervals alongside KPI predictions, supporting risk-aware decision-making in high-precision semiconductor manufacturing.
- **Applied Explainable AI (XAI) methods** such as GradCAM and LIME to enhance model interpretability, aiding validation, debugging, and building trust with domain experts in optical metrology.

FABHEADS AUTOMATION PRIVATE LIMITED – CHENNAI, INDIA

Business or Sector Manufacturing | **Website** www.fabheads.in

Links https://www.youtube.com/watch?v=wbnzM_H04qM&t=14s | https://www.youtube.com/watch?v=7p62mFohdr0&t=7s

CHIEF PRODUCT OFFICER – 01/09/2019 – 12/07/2023

- **Led a cross-functional R&D team of 10+ engineers and scientists**, overseeing the development of advanced manufacturing technologies and aligning product strategy with business objectives.
- **Invented and co-developed 7 granted patents** in composites and additive manufacturing, including the creation of **Asia's and India's first Carbon Fiber 3D Printer** and **India's fastest pellet-based 3D printer**.
- **Pioneered India's first AFP (Automated Fiber Placement) Winding Machine** and **developed a proprietary slicing engine** that enabled continuous fiber 3D printing, unlocking new capabilities in structural part fabrication.
- **Defined product roadmaps, secured R&D proposals, and managed IP portfolios**, driving innovation while ensuring protection of key technologies through patents, software copyrights, and trademarks.

FABHEADS AUTOMATION PRIVATE LIMITED – CHENNAI, INDIA

Business or Sector Manufacturing | **Department** R&D

HEAD, ENGINEERING – 01/07/2017 – 01/09/2019

- **Led a cross-functional team of 6 engineers** (mechanical, electronics, and software) to rapidly prototype and deliver MVPs for industrial 3D printers, balancing hardware-software integration with iterative development cycles.
- **Directed technical design reviews (PDRs/CDRs)** to ensure system-level alignment, design feasibility, and risk mitigation across mechanical, electrical, and embedded software domains.

- **Established and maintained engineering documentation workflows**, including version-controlled BOMs, CAD-linked process docs, and structured assembly protocols for traceability and collaboration.
- **Collaborated closely with manufacturers and vendors**, optimizing part designs for manufacturability and cost-efficiency, while ensuring smooth handoff between engineering and production teams.

PROJECTS

01/09/2023 – CURRENT
Artificial Intelligence Motor Imagery

- Leading the Machine Learning team at Serpentine AI (TUe Student Team) to develop a robust deep learning model to predict EEG Motor imagery data in real-time.
- Created custom datasets and transformer-based models and achieved 86% accuracy on the Physionet Motor Imagery Dataset
- Use of novel data augmentation techniques such as channel scrambling and data stitching to improve accuracy with fine-tuning on real data.

01/03/2024 – CURRENT
EmbedAnything

- Author of the library with over **500 stars** and **200,000 downloads.**
- EmbedAnything is a lightning fast embedding creation library built in Rust and Python.
- Implemented re-ranking, late chunking and vector streaming features.
- Contributed three models to HuggingFace for use in EmbedAnything.

Link https://github.com/StarlightSearch/EmbedAnything

01/02/2025 – CURRENT
Lumo Agent

- An open-source agentic library and an agent API server that can be used to make intelligent chat applications.
- Implements CodeAgent architecture where the agent writes Python code to execute tool calls.
- Implements tracing for agent observability and tracking with OpenTelemetry.

Link https://github.com/StarlightSearch/lumo

SKILLS

Programming Languages

Rust | Python | Typescript

Frameworks and Libraries

PyTorch | Tensorflow | Langchain | LlamaIndex | OpenTelemetry | Pandas

Software Development

Docker | Git | CLI | Experiment Tracking

Web Development

Sveltekit | FastAPI | Firebase

LANGUAGE SKILLS

Mother tongue(s): **MARATHI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
HINDI	C1	B2	C1	C1	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user